CRYPTO MEDIUM-HARD

Hensel Garden

Cryptosystem Overview

Authored by 5h1kh4r. Files: `chall.py` and `data.json`. A 128-bit secret integer seed $s \in [0, 2^{128}-1]$ is shifted and squared modulo two prime powers:

- $(s + 87)^2 \equiv sq_1 \pmod{2^{72}}$
- $(s + 143)^2 \equiv sq_2 \pmod{3^{51}}$

Total product modulus: $M = 2^{72} \times 3^{51} \approx 2^{152.83} > 2^{128}$

Because $M > 2^{128}$, the CRT reconstruction over $\mathbb{Z}/M\mathbb{Z}$ has a unique integer root lying in $[0, 2^{128}-1]$.

Root Lifting via Hensel's Lemma

1. **Modulo 2^{72} (2-adic square roots):**

- For an odd square modulo 2^k ($k \ge 3$), there are exactly 4 square roots.
- Base roots modulo 8: $r^2 \equiv sq_1 \pmod 8$.
- Inductive lift: If $r_k^2 \equiv sq_1 \pmod{2^k}$, then the root mod 2^{k+1} is either r_k or $r_k + 2^{k-1}$.
- Lifting iteratively to $k=72$ yields 4 distinct candidates for $(s + 87) \pmod{2^{72}}$.

2. **Modulo 3^{51} (Hensel's Lemma for Odd Primes):**

- Function: $f(x) = x^2 - sq_2 = 0$.
- Base roots modulo 3: $x_0^2 \equiv sq_2 \pmod 3$ (2 roots).
- Derivative: $f'(x) = 2x \not\equiv 0 \pmod 3$.
- Unique Hensel lift:

$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} \pmod{3^{k+1}}$

- Lifting to $k=51$ gives 2 distinct roots for $(s + 143) \pmod{3^{51}}$.

CRT Combination & Flag Recovery

Combining the 4 roots from $\mathbb{Z}/2^{72}\mathbb{Z}$ with the 2 roots from $\mathbb{Z}/3^{51}\mathbb{Z}$ produces $4 \times 2 = 8$ candidates modulo M.

```
[*] Modulus 1: 2^72 = 4722366482869645213696
[*] Modulus 2: 3^51 = 215369396307555776631074 7
[*] Combined Modulus M = 2^72 * 3^51 (153 bits > 128-bit secret)
[*] Lifting 4 roots mod 2^72 and 2 roots mod 3^51 via Hensels Lemma...
[+] Total CRT candidates = 4 x 2 = 8
[+] Filtering by bound 0 <= s < 2^128 uniquely identifies true root!

~
❯
```

Hensel Lifting Modulus Analysis

Testing the bound condition $0 \le s < 2^{128}$ filters out the 7 invalid CRT residues, uniquely recovering the seed: `s = 282779946134366997745983790016833981964`

Solution Script: Hensel_Garden.py (included in attached solution archive)

Execution Output

```
$ python3 solve.py
```

```
~/SecLeaf writeups/repo_clone/SecLeafCTF/CRYPTO/Hensel_Garden main
❯ python3 solve.py
[+] Recovered 128-bit seed: 282779946134366997745983790016833981964
[+] Flag: SecLeaf{r00ts_gr0w_through_pr1me_p0w3rs}

~/SecLeaf writeups/repo_clone/SecLeafCTF/CRYPTO/Hensel_Garden main
❯
```

Flag Execution Proof

Solution Script: Hensel_Garden.py (included in attached solution archive)

Flag:

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RECOVERED FLAG **SecLeaf{r00ts_gr0w_through_pr1me_p0w3rs}**